

Convexity, end to end

An indexed summary of the tail-hedge investigation — the barbell, the curveball, funded vs paid convexity, carry, and the diversified tail hedge. Nine sketches, their verdicts, the durable laws they establish, and the one idea that earned its way to a live test.

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RESEARCH THREAD — THOUGHT EXERCISES, NOT DEPLOYED

Every result below is a research sketch on ETF proxies, not a strategy managing capital, with one stated exception (multi-horizon trend, now in a paper A/B). All are reproducible under [scripts/research/](#). Read the *shapes and directions*, not the point estimates — tail-strategy backtests are dominated by which crises fell in the window, and this one holds only a single sharp crash (COVID) and a single inflation bear (2022).

SUMMARY IN ONE PARAGRAPH

We asked whether convex tail protection could be added to the Spine cheaply. It cannot — not for free. The investigation established a clean map: **convexity is either free (funded by the momentum premium, via trend — it protects *sustained* crises and improves return) or paid (funded by decay, via long-vol/short-carry — it protects *sharp* crises but bleeds, and cannot be timed)**. Along the way: a paid overlay on the Spine *hurts* it; making the trend sleeve multi-horizon is a clean win and the one idea we graduated; carry is a real but negative-skew premium that damages convex sleeves; and the best tail hedge is *diversified across crisis regimes* (deflation vs inflation), which are anti-correlated. The single durable lesson: you don't *time* the tail or *optimize* it — you budget it, diversify what it insures against, and let the free convexity in trend do the rest.

1 The index

Nine sketches, in the order they were run. Each is a self-contained script under [scripts/research/](#).

#	SKETCH	THE QUESTION	THE VERDICT
1	barbell_sketch	Does a Taleb barbell (90% safe / 10% convex) work, and its reversal — the "curveball" (90% convex)?	Barbell = paid insurance : bleeds ~3%/yr, skew +2.1, positive in every crisis (+19% COVID). Curveball = a dark horse: ruin if compounded (−44%/yr) but +237% in COVID on its 10% — hold it, let it bleed, never time it.
2	spine_tail_overlay	Does a vol-aware VIXY tail overlay improve the Spine?	No — it hurts. The Spine's trend sleeve is already a crisis hedge; VIXY carry is a drawdown machine in calm; "buy when vol is cheap" over-carries the bleed. Paid convexity on a book that already has free convexity is pure cost.
3	convex_trend	Can the trend sleeve be made more convex — for free?	Yes — multi-horizon (3/6/12mo). Sharpe 1.04→1.16, drawdown −8.9→−7.0%, more convex, no theta. The obvious "convex response function" (strength ²) backfires — it levers calm uptrends and flattens skew. Convexity must be in the payoff, not the sizing.
4	multi_horizon_split	Do multi-horizon trend and split-universe breadth stack?	Substitutes, not complements. On Sharpe they don't add (both ~1.1); the combination only wins on tail-shape. Multi-horizon on the plain 6-asset universe is the Sharpe winner and the simplest — the drop-in upgrade.
5	crash_and_curveball	Can a faster/break trend catch a sharp crash? Can the curveball be rescued?	No, and no. Faster trend fights reversal and whipsaws the V-recovery (COVID still ~0). Every rule that trims the curveball's bleed also removes its payoff (vol-gating: +237%→0%). Establishes the free-vs-paid capstone .
6	carry_sleeve	Does adding carry as a third premium diversify (the "Holy Grail")?	Uncorrelated but doesn't pay. Carry's P&L is genuinely uncorrelated to base (−0.02) / trend (−0.18), but it lost money (−2%/yr in this window), has negative skew (−2.5), and lowers the blend's Sharpe (1.16→0.96). Diversification is necessary, not sufficient.

#	SKETCH	THE QUESTION	THE VERDICT
7	carry_curveball	Can carry's income <i>fund</i> the curveball's bleed (carry as the barbell's earning sleeve)?	No — it eats the payoff. Carry crashes in the same disasters the curveball pays (COVID: carry -10% vs bills ~0), shrinking the net (+17%→+6%); in a slow bear both lose. The safe sleeve must be <i>truly</i> safe — bills doing nothing is the feature.
8	inverse_carry	What about <i>shorting</i> carry — short high-carry instead of long?	A real second hedge. Inverting flips skew -2.1→+2.1; it catches the rate/credit crisis (2022 +5%) that VIXY misses. Caveat: its positive return here is a 2022 window artifact, it's milder than VIXY, and +0.43 correlated. Tail hedges can be diversified across disaster <i>types</i> .
9	tail_hedge_diversification	Can we hedge every crisis type with one budgeted sliver?	Yes — hedge the two regimes. Deflation hedges (VIXY, short-credit) and inflation hedges (short-duration, long-commodities) are anti-correlated (-0.3), so a blended sliver is positive in every crisis, at a lower bleed than any single hedge — but breadth dilutes explosiveness (COVID +3% vs VIXY +19%).

2 The durable laws

Five conclusions survive the whole thread. They are directional and structural — the kind of finding robust to the small-sample caveats, unlike any single return number.

I. Convexity is free or paid — never both cheap and fast.

Free convexity (trend / time-series momentum, funded by the momentum premium) protects *sustained* crises like 2022 and *improves* return. **Paid convexity** (long-vol, short-carry, funded by decay) protects *sharp* crises like COVID but *bleeds*. No instrument tested delivers cheap, well-timed, sharp-crash protection — that is the shape of the problem, not a search we abandoned.

II. Timing the tail removes the tail.

Every rule that reduced a paid hedge's bleed also removed its crash payoff, because the bleed and the convexity are the same thing (being positioned in the decaying instrument at all times). Vol-gating the curveball took COVID from +237% to 0%. You budget paid convexity as an un-timed premium — you do not optimize *when* to hold it.

III. Diversification is necessary, but not sufficient.

An uncorrelated return stream only helps the ensemble if it is *positive-expectancy*. Carry was genuinely uncorrelated (-0.02 to base) yet lowered the blended Sharpe, because its own expectancy (in this proxy/window) was negative and its skew was the wrong sign. Low correlation is a filter, not a guarantee.

IV. A hedge's safe sleeve must be inert in a crisis.

Replacing the barbell's bills with a carry sleeve failed precisely because carry *does something* in a crisis — it crashes — eating the convex payoff. Bills doing nothing in a disaster is not a weakness; it is the entire point. Anything with hidden crisis-beta (credit, duration, "reach for yield") is not a safe sleeve, however high its calm-market yield.

V. There are two crisis regimes, and they need opposite hedges.

Deflationary crashes (equities down, bonds up, vol spikes) want VIXY and short-credit; inflationary crashes (equities *and* bonds down, like 2022) want short-duration and long-commodities. The two hedge-clusters are *anti-correlated* (-0.3), so a small sliver spread across both has no crisis blind spot and a lower net premium than any single hedge. Diversify the tail hedge across *what it insures against*, not just how much of it you hold.

3 The two-regime map (Law V, in numbers)

Crisis capture by hedge, showing the regime split — each single hedge has a blind spot; the blend does not.

HEDGE	REGIME IT COVERS	COVID '20 (DEFLATION)	2022 (INFLATION)	CALM BLEED
VIXY (long vol)	deflationary vol spike	+19%	+1%	−7%/yr
short-credit (short HYG)	deflationary credit crisis	+18%	+18%	−8%/yr
short-duration (short TLT)	inflationary rate shock	−9%	+24%	−2%/yr
long-commodities (DBC)	inflation	−14%	+13%	+5%/yr*
diversified sliver	both regimes	+3%	+14%	−3%/yr

* long-commodities' positive carry is a 2016–2026 window artifact (the inflation era); over a normal window it is commodity beta, not a hedge.

4 What graduated, and what was declined

Graduated to a live test

Multi-horizon trend (3/6/12-month agreement) — the one strict improvement (higher Sharpe, shallower drawdown, more convex, no cost, a standard untuned technique). Wired into `spine.jl` behind `BB_TREND_MODE` and now running a paper A/B on a dedicated third account (`single-sign` vs `split-sign` vs `single-multi`), pending OOS/purged-CV before any funded path.

Reviewed, understood, declined

Paid tail overlays on the Spine (redundant with its free convexity); the curveball as anything but an inert-funded, un-timed insurance sliver; the "convex response function" (backfires); carry as a return sleeve (uncorrelated but negative-expectancy) and as a barbell base (crashes when the hedge pays). Each is on record with its reason, so none needs re-litigating.

Standing, if a sharp-crash hedge is ever wanted

A small, budgeted, *diversified* tail sliver — spread across the two crisis regimes (VIXY + short-credit for deflation; short-duration + long-commodities for inflation) — held always, never timed, sized as insurance you have decided in advance you can lose. That is the honest form of the "dark horse."

5 Meta-caveats

Three limits apply to every number above and should temper any reading of the returns (the *directions* are far more trustworthy than the magnitudes):

1. One of each crisis. The 2016–2026 window contains a single sharp crash (COVID) and a single inflation bear (2022). Tail results are dominated by these two events; a different decade would move every crisis-capture and bleed figure.

2. ETF proxies, not the real instruments. Carry is proxied by income yield (which conflates it with credit/duration beta); the long-vol hedge is VIX futures (VIXY), not options; commodity carry is muddled by collateral yield. A true cross-asset version needs futures/rates data outside our reach.

3. Window-artifact returns. Several "positive" results (inverse carry, short-duration, long-commodities) earned only because 2022 fell in the sample. They are shorting or tilting against premia that are positive in the long run; do not extrapolate their carry.

Reproducible sketches (all under `scripts/research/`): `barbell_sketch.py`, `spine_tail_overlay_sketch.py`, `convex_trend_sketch.py`, `multi_horizon_split_sketch.py`, `crash_and_curveball_sketch.py`, `carry_sleeve_sketch.py`, `carry_curveball_sketch.py`, `inverse_carry_sketch.py`, `tail_hedge_diversification_sketch.py`. Companion notes: `docs/barbell_note.pdf`, `docs/research_frontier.html`. Validated strategy: `docs/whitepaper.html` / `FINANCIAL_METHODS.md`. Decisions: `docs/CANONICAL_ARCHITECTURE.md`.

Research summary. Describes thought exercises, not strategies managing capital; not investment advice, an offer, or a solicitation. All figures are backtested/simulated on ETF proxies and are not indicative of future results; tail-strategy backtests are especially sample-dependent. Trading carries substantial risk, including loss of principal. © 2026 Carter Warrens · MIT-licensed.